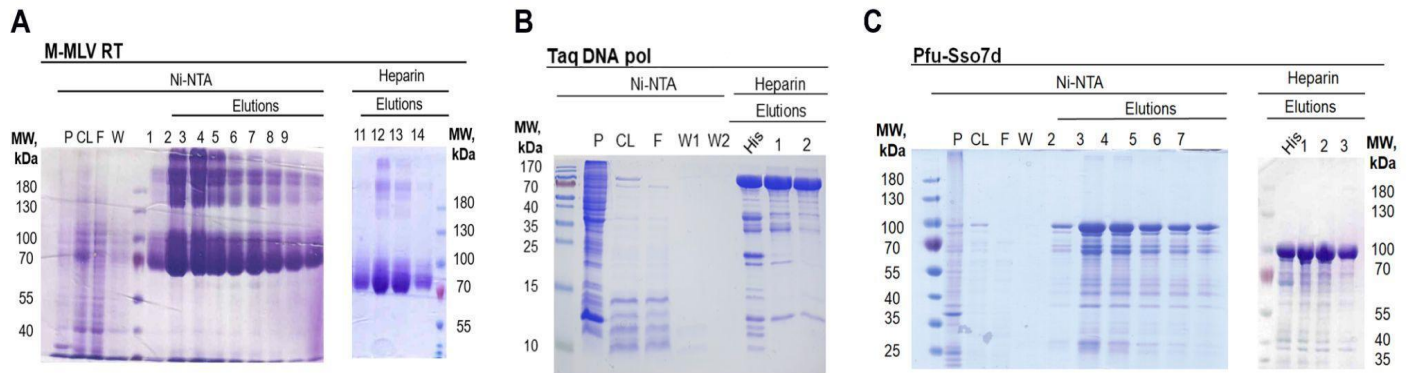
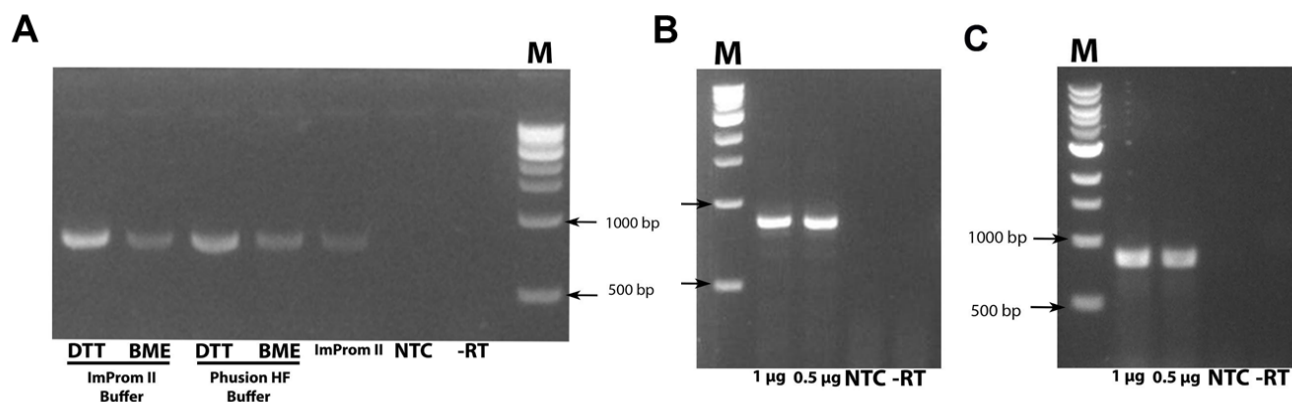


Supplementary Information



Supplemental Figure 1. Purification of M-MLV RT, Taq DNA pol and Pfu-Sso7d enzymes. SDS-PAGE of M-MLV RT (**A**) Taq DNA pol (**B**) and Pfu-Sso7d (**C**) purification. Samples were prepared by volume; 3 μ L for pellet (P), clarified lysate (CL), and flowthrough (F); 12 μ L for wash (W) and elution samples from the Ni-NTA and heparin purifications. The observed molecular weight of the purified proteins matches the one predicted based on their amino acid sequences: 80 kDa for M-MLV RT, 96 kDa for Taq DNA pol, and 100 kDa for Pfu-Sso7d. The His lane in the heparin purification panel in (B) and (C) corresponds to a pooled Ni-NTA purified protein sample before heparin purification. SDS-PAGE gels in (A) and (C) were made at 10% polyacrylamide (PA), whereas 12% PA was used for (B).



Supplemental Figure 2. Homebrew M-MLV RT, Taq DNA pol and Pfu-Sso7d activity testing. (A) Determination of M-MLV RT functionality in a conventional RT-PCR. A Two-Step RT-PCR was carried out using M-MLV RT for cDNA synthesis and Pfu-Sso7d for DNA amplification. ImProm-II 5X Reaction Buffer (Promega) and 5X Phusion HF Buffer (Thermo Scientific) were used. Additionally, dithiothreitol (DTT) and β -mercaptoethanol (BME) were tested as reducing agents. A positive control using ImProm II-RT in the 5X Reaction Buffer (Promega) was added. **(B)** One-Step RT-PCR using M-MLV RT for cDNA synthesis and Taq DNA pol for DNA amplification in a homemade 5X reaction buffer (composition described in Material & Methods) reaction buffer. Different amounts of total RNA treated with DNase (1 and 0.5 μ g) were used as a template. **(C)** One-Step RT-PCR using M-MLV RT for cDNA synthesis and Pfu-Sso7d for DNA amplification in the same conditions described in (B). The expected band size (~853 bp fragment), was obtained in all the samples. **NTC**: No template control. **-RT**: No RT control. **M**: DNA ladder 1 Kb (NEB).

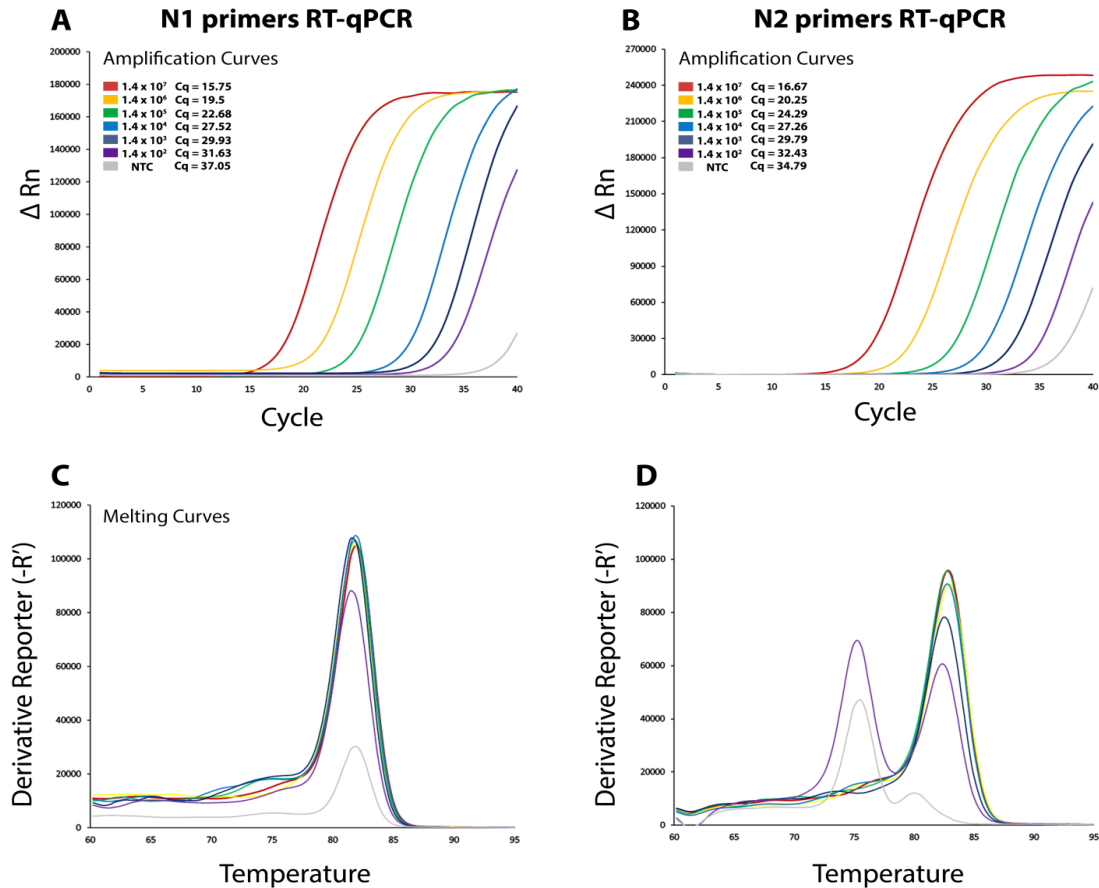
Supplemental Table 1. Dye-based RT-qPCR reaction mix using M-MLV RT and Pfu-Sso7d.

One-Step RT-qPCR (M-MLV/Pfu)	Volume (μL)	Final Concentration
RNA sample	5	-
10 μ M Forward PCR primer	0.6	300 nM
10 μ M Reverse PCR primer	0.6	300 nM
10 mM dNTPs	0.8	400 nM
5X Homemade Buffer	4	1X
100 mM DTT	2	10 mM
20X EvaGreen	1	1X
Pfu Ss07d (0.6 mg/mL)	0.5	15 ng/ μ L
M-MLV RT (0.02 mg/mL)	0.125	0.13 ng/ μ L
Nuclease-Free Water	5.375	-
Total Reaction Volume	20	

Supplemental Table 2. Dye-based One-Step RT-qPCR cycling conditions.

Step	Temperature ($^{\circ}$C)	Duration
1	50	15 min
2	98	2 min
Repeat steps 3 and 4 by 40 cycles		
3	98	10 s
4*	55	10 s
Melt Curve		
5	95	15 s
6	60	1 min
7	95	15 s

*In this step fluorescence signal acquisition occurs (EvaGreen)



Supplemental Figure 3: CDC SARS-CoV-2 N1 and N2 One-Step RT-qPCR assays performed with synthetic RNA using homebrew M-MLV RT and Pfu-Sso7d. (A-B) Representative amplification curves using EvaGreen as DNA intercalating dye. Each curve represents a specific dilution of SARS-CoV-2 synthetic N RNA used as template: 1.4×10^7 copies approximately (red line), 1.4×10^6 (yellow line), 1.4×10^5 (green line), 1.4×10^4 (light blue line), 1.4×10^3 (blue line), 1.4×10^2 (purple line) and no template control (**NTC**, gray line). Characteristic C_q values are indicated on the upper left side of each panel in order to evaluate if amplification curves correspond to single amplicons (peaks); N1 and N2 melting curve analyses are shown in panels **C** and **D**, respectively.

Supplemental Table 3. Comparative Cq data between a commercial RT-qPCR kit and an Open RT-qPCR method based on homebrew M-MLV RT and Taq DNA pol.

# Sample	ID	Commercial Kit				Open RT-qPCR (M-MLV/Taq)			
		N1	N2	RNAse P	Report	N1	N2	RNAse P	Report
1	1215-02 3 (+)	13.28	14.01	26.42	Positive	14.53	14.97	26.08	Positive
2	1229-05 5 (+)	14.40	14.08	28.31	Positive	15.97	16.47	28.09	Positive
3	1228-00 1 (+)	18.35	19.51	27.17	Positive	19.43	20.22	27.64	Positive
4	1228-04 8 (+)	18.46	17.95	25.41	Positive	19.36	18.26	26.31	Positive
5	1230-14 0 (+)	18.78	18.97	25.66	Positive	19.68	19.1	29.15	Positive
6	1228-04 7 (+)	19.57	19.54	30.80	Positive	17.67	18.29	28.00	Positive
7	1228-06 6 (+)	19.98	19.76	25.05	Positive	20.49	19.97	25.35	Positive
8	1230-02 4 (+)	20.27	20.62	28.46	Positive	20.72	21.42	27.13	Positive
9	1229-03 4 (+)	20.41	20.91	31.95	Positive	20.72	20.76	31.40	Positive
10	1228-03 5 (+)	21.05	20.20	31.81	Positive	21.88	22.64	33.91	Positive
11	1228-09 6 (+)	23.55	23.29	25.71	Positive	25.75	23.52	25.70	Positive
12	1230-09 6 (+)	23.60	23.12	27.37	Positive	23.58	23.72	26.47	Positive
13	1230-14 2 (+)	24.20	24.84	31.77	Positive	25.38	25.76	30.61	Positive
14	1229-05 4 (+)	24.24	24.35	27.38	Positive	24.88	26.46	27.68	Positive
15	1228-04 0 (+)	25.73	25.39	27.73	Positive	26.82	25.83	27.95	Positive
16	1230-10 1 (+)	26.10	25.91	30.24	Positive	27.94	26.28	33.47	Positive
17	1228-02 1 (+)	26.42	26.25	30.22	Positive	27.03	27.35	30.14	Positive
18	1228-10 2 (+)	31.50	31.49	26.74	Positive	32.89	34.61	26.33	Positive
19	1230-04 4 (+)	32.14	31.84	29.17	Positive	34.78	32.34	29.24	Positive
20	1229-03 3 (+)	ND	ND	29.52	Negative	ND	ND	29.59	Negative
21	0421-0P 1 (-)	ND	ND	25.29	Negative	ND	ND	26.90	Negative
22	0421-0P 2 (-)	ND	ND	26.73	Negative	ND	ND	26.56	Negative
23	0414-0P 3 (-)	ND	ND	28.39	Negative	ND	ND	29.70	Negative
24	0414-0P 4 (-)	ND	ND	27.72	Negative	ND	ND	26.91	Negative
25	0414-0P 5 (-)	ND	ND	27.57	Negative	ND	ND	27.60	Negative

26	<i>0414-OP 6 (-)</i>	ND	ND	28.29	Negative	ND	ND	27.08	Negative
27	<i>0414-OP 7 (-)</i>	ND	ND	26.98	Negative	ND	ND	26.87	Negative
28	<i>0414-OP 8 (-)</i>	ND	ND	27.78	Negative	ND	ND	26.03	Negative
29	<i>0414-OP 9 (-)</i>	ND	ND	25.34	Negative	ND	ND	26.26	Negative
30	<i>0414-P1 0 (-)</i>	ND	ND	29.45	Negative	ND	ND	28.64	Negative
31	<i>0414-P1 1 (-)</i>	ND	ND	28.95	Negative	ND	ND	28.88	Negative
32	<i>0414-P1 2 (-)</i>	ND	ND	26.98	Negative	ND	ND	27.07	Negative
33	<i>0414-P1 3 (-)</i>	ND	ND	26.54	Negative	ND	ND	26.11	Negative
34	<i>0414-P1 4 (-)</i>	ND	ND	25.16	Negative	ND	ND	26.56	Negative
35	<i>0414-P1 5 (-)</i>	ND	ND	27.11	Negative	ND	ND	27.90	Negative
36	<i>0414-P1 6 (-)</i>	ND	ND	29.74	Negative	ND	ND	28.50	Negative
37	<i>0414-P1 7 (-)</i>	ND	ND	25.85	Negative	ND	ND	28.14	Negative
38	<i>0414-P1 8 (-)</i>	ND	ND	27.74	Negative	ND	ND	29.20	Negative
39	<i>0414-P1 9 (-)</i>	ND	ND	25.45	Negative	ND	ND	26.87	Negative
40	<i>0414-P2 0 (-)</i>	ND	ND	28.67	Negative	ND	ND	27.45	Negative

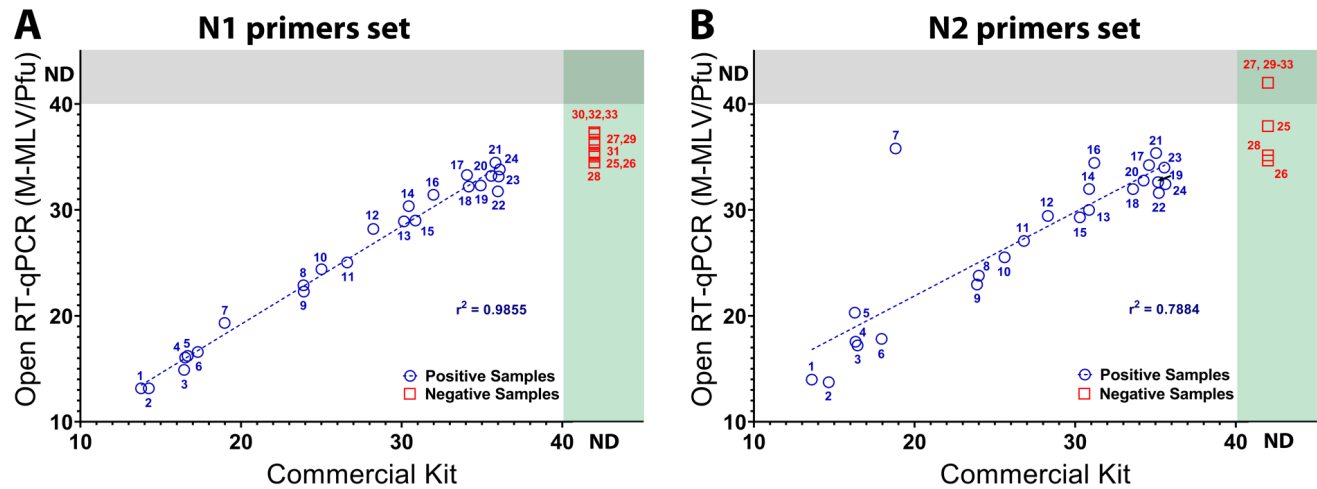
Comparative Cq data for the TaqPath One-Step RT-qPCR kit and the probe-based open RT-qPCR reaction mix. Assigned sample number (# Sample) and clinical sample identifier (ID) are displayed. The clinical reports of the samples before they were re-tested by the two kits are also indicated in parentheses. The sample whose report was altered is denoted in bold. **(-)**: negative samples, **(+)**: positive samples, **ND**: non-detected.

Supplemental Table 4. Comparative Cq data between a commercial RT-qPCR kit and an Open RT-qPCR method based on homebrew M-MLV RT and Pfu-Sso7d.

# Sample	ID	Commercial Kit				Open RT-qPCR (M-MLV/Pfu)			
		N1	N2	RNAse P	Report	N1	N2	RNAse P	Report
1	4896 (+)	13.78	13.60	27.13	Positive	13.16	13.98	26.79	Positive
2	5015 (+)	14.26	14.66	26.60	Positive	13.16	13.73	27.10	Positive
3	5207 (+)	16.46	16.33	30.67	Positive	14.9	17.56	29.76	Positive
4	5342 (+)	16.53	16.45	28.95	Positive	16.06	17.21	31.58	Positive
5	5426 (+)	16.68	16.28	30.62	Positive	16.22	20.3	29.25	Positive
6	4907 (+)	17.31	17.95	28.18	Positive	16.59	17.83	27.58	Positive
7	5058 (+)	18.98	18.82	26.88	Positive	19.33	35.79	27.02	Positive
8	5062 (+)	23.89	24.00	25.90	Positive	22.88	23.79	25.42	Positive
9	5085 (+)	23.91	23.89	27.48	Positive	22.29	22.96	27.62	Positive
10	5438 (+)	25.01	25.61	31.28	Positive	24.4	25.52	30.70	Positive
11	4872 (+)	26.61	26.81	31.19	Positive	25.04	27.07	30.88	Positive
12	4738 (+)	28.24	28.30	27.54	Positive	28.2	29.43	29.88	Positive
13	5472 (+)	30.15	30.86	29.62	Positive	28.91	29.99	29.09	Positive
14	4883 (+)	30.44	30.87	28.25	Positive	30.35	31.97	28.89	Positive
15	5181 (+)	30.86	30.29	29.54	Positive	28.99	29.29	30.66	Positive
16	4798 (+)	31.98	31.19	27.92	Positive	31.43	34.43	27.50	Positive
17	5179 (+)	34.07	34.60	31.58	Positive	33.29	34.23	30.51	Positive
18	4727 (+)	34.18	33.60	31.14	Positive	32.2	31.96	31.95	Positive
19	4978 (+)	34.92	35.16	28.83	Positive	32.3	32.62	27.29	Positive
20	5384 (+)	35.58	34.26	26.10	Positive	33.21	32.77	26.86	Positive
21	2528 (+)	35.84	35.04	26.10	Positive	34.45	35.36	26.40	Positive
22	5198 (+)	35.99	35.21	31.66	Positive	31.75	31.59	30.72	Positive
23	5093 (+)	36.04	35.55	26.03	Positive	33.15	33.99	26.11	Positive
24	5174 (+)	36.11	35.62	25.57	Positive	33.82	32.42	26.44	Positive

25	2514 (-)	ND	ND	26.36	Negative	35.03	37.91	26.42	Positive
26	2529 (-)	ND	ND	27.40	Negative	35.2	34.66	25.32	Positive
27	2486 (-)	ND	ND	28.96	Negative	36.62	ND	26.82	Inconclusive
28	2513 (-)	ND	ND	27.26	Negative	34.47	35.15	26.63	Positive
29	2489 (-)	ND	ND	25.33	Negative	36.38	ND	25.25	Inconclusive
30	2487 (-)	ND	ND	27.44	Negative	37.17	ND	26.99	Inconclusive
31	2477 (-)	ND	ND	28.85	Negative	35.34	ND	28.95	Inconclusive
32	2473 (-)	ND	ND	25.90	Negative	37.32	ND	26.24	Inconclusive
33	2517 (-)	ND	ND	28.32	Negative	37.32	ND	26.64	Inconclusive

Comparative Cq data for the TaqPath One-Step RT-qPCR kit and the dye-based open RT-qPCR reaction mix. Assigned sample number (# Sample) and clinical sample identifier (ID) are displayed. The clinical reports of the samples before they were re-tested by the two kits are also indicated in parentheses. The samples whose reports were altered are denoted in bold. (-): negative samples, (+): positive samples, **ND**: non-detected.



Supplemental Figure 4: Negative samples cannot be successfully identified using the dye-based Open One-Step RT-qPCR reaction mix (M-MLV RT and Pfu-Sso7d). Scatterplot of the Cq values of positive (blue circles) and negative (red squares) samples obtained by the commercial and open RT-qPCR reaction master mixes using the N1 and N2 primer sets (**A** and **B**, respectively). The numbers displayed in each sample match those displayed in Supplemental Table 3 (# Sample). If a Cq value was not detected in the sample, it appears in the ND area of the graph depending on whether this occurred in the commercial kit (green rectangle), the open dye-based kit (gray rectangle), or both (intersections between the rectangles). For each combination of primers, the linear trend of the positive samples is shown (blue dotted line) along with the corresponding value of r^2 . **ND**: non-detected